

Multiple Choice

1. **Answer:** D

Options: A) Eu; B) Gd; C) Am; D) Cu

Note: Correct option: D - Cu

2. **Answer:** A

Options: A) 10 ml; B) 30 ml; C) 55 ml; D) 40 ml; E) 25 ml

Note: Correct option: A - 10 ml

3. **Answer:** A

Options: A) $1.6 \times 10^{-3}F$; B) $1.5 \times 10^{-3}F$; C) $1.4 \times 10^{-3}F$; D) $0.8 \times 10^{-3}F$; E) $0.9 \times 10^{-3}F$

Note: Correct option : A – $1.6 \times 10^{-3}F$

3. **Answer:** E

Options: A) .1002 : .3002 : .4002; B) .3582 : .4582 : .0582; C) .2582 : .4998 : .3092;

D) .1582 : .6008 : .2092; E) .1582 : .4998 : .1092

Note: Correct option: E - .1582 : .4998 : .1092

4. **Answer:** B

Options: A) $He_2^{\oplus}$ is the most stable among He , He_2 , and $He_2^{\oplus}$; B) $He_2^{\oplus}$ is more stable than He_2

Note: Correct option : B – $He_2^{\oplus}$ is more stable than He_2 but less stable than He

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '48.359 eV'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '0.000549 amu'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 2.5 gallons. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** 3000 ft^3 . Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.